Suntracker

User Manual and Commissioning/Service Instructions

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Status: short appendix to the project documentation

This document supplements the project documentation with two missing sections: the user manual and the commissioning/service instructions. It does not replace the main documentation, schematics, PCB files, source code, BOM, simulations or test results.

1 Scope and Intended Use

Suntracker is a prototype autonomous solar tracker with a solar powerbank function and a simplified MPPT algorithm. The device positions the panel toward more favorable lighting, measures panel parameters, controls the actuators and reduces energy consumption by entering sleep mode.

The prototype is intended for educational and validation purposes. It is not a finished device designed for long-term, unattended outdoor operation and is not protected against moisture.

2 User Manual

2.1 Safety Rules

- Do not operate the prototype in rain, at high humidity or on a conductive surface.
- Do not touch the mechanism while the motors are moving, and do not block the panel by hand.
- Do not use the system if the cell is swollen, hot, mechanically damaged or has damaged insulation.
- Do not short-circuit the cell output, panels, 5 V line or 3.3 V line.
- Perform the first powerbank function tests with a test load, not an expensive device.

2.2 Preparation for Operation

1. Place the device on a stable surface with enough space for the panel to move freely.
2. Make sure that the wires do not enter the working area of the arm, lead screw or servo.
3. Point the panel toward the light or place a light source in front of the device.
4. Before connecting a USB load, use a multimeter to verify that the output voltage is approximately 5 V.

2.3 Startup and Normal Operation

1. Press the `POWER_ON` power/wake-up button.
2. After startup, the system should activate the required power sections and begin the panel unfolding or positioning sequence.
3. In `Harvest_update` mode, the device searches for a more favorable panel position using the photoresistors and power measurement.
4. In `Harvest` mode, the system keeps the panel in its operating position and performs MPPT adjustments.
5. Under low-light conditions, the system may limit movement and enter a power-saving mode.

Correct operation means short, controlled movements of the mechanism, no microcontroller resets, and no noticeable heating of the wires, drivers or converters.

2.4 Shutdown, Folding and Storage

1. Press the power button.
2. Wait for the mechanism to complete the folding movement.
3. Do not disconnect the cell while the motors are moving unless an emergency occurs.

2.5 Emergency Conditions

Stop the test immediately if smoke, the smell of overheated electronics, excessive cell heating, continuous system resets, a blocked mechanism, sparking, or a significant deviation of the 5 V or 3.3 V rail from its nominal value occurs.

3 Commissioning and Service Instructions

3.1 Required Tools

Multimeter	checking for short circuits, polarity, and the 3.3 V, 5 V, cell and panel voltages.
Laboratory power supply	first power-up with current limiting.
ST-Link / SWD	programming the STM32L0 microcontroller.
Computer with an IDE	compiling and uploading the source code.
Lamp / light source	testing the photoresistors and tracking algorithm.
Test load	testing the 5 V output and powerbank function.

3.2 Inspection Before First Power-Up

1. Compare the assembled PCB with the BOM and schematic.
2. Check the polarity of the cell, panels, diodes and electrolytic capacitors.
3. Measure the resistance between `VBAT-GND`, `5V-GND`, `3V3-GND` and `PANEL+-PANEL-`. There should be no short circuit.
4. Check that the mechanism moves freely within its intended range.
5. Check for collisions between PCB components and the motor, lead screw, servo and arm.

3.3 First Electronics Startup

1. Disconnect the motors, servo and USB load.
2. Connect power using a laboratory power supply with current limiting, or use a tested 1S cell.
3. Check the stability of the 3.3 V rail. Then check the 5 V rail if it is active.
4. Check that the microcontroller is not resetting repeatedly.
5. Use a debugger to verify I2C communication with the INA219 and MCP40xx devices.

3.4 Microcontroller Programming

1. Connect `GND`, `SWDIO`, `SWCLK` and, optionally, `NRST` from the ST-Link programmer.
2. Select the board/microcontroller variant that matches the STM32L0 used in the device.
3. Make sure that the environment contains the required libraries: STM32, Wire/I2C, timers, STM32LowPower, STM32RTC, INA219 and MCP40xx.
4. Compile and upload the firmware. After uploading, reset the system and check its response to the `POWER_ON` button.

3.5 Actuator and Sensor Startup

Connect the actuators in stages. First test one DC motor in both directions, then the second motor, followed by the servo over a limited range. If the direction of movement is opposite to the expected direction, swap the motor wires or reverse the logic in the control function.

Test the photoresistors under uniform illumination and by covering the sensors one at a time. Illuminating one side should change the indicated direction. If the directions are shifted, correct the pin mapping or the description of the sensor orientation.

3.6 MPPT and Sleep Mode Test

For the MPPT test, check the voltage and current readings from the INA219 and the system response to changes in illumination or load. Changing the MCP40xx setting should affect the operating conditions of the charging circuit. If the algorithm oscillates too much, reduce the adjustment step or increase the measurement settling time.

For sleep mode, confirm that the System_Sleep() function switches off unnecessary power sections, places the H-bridge inputs in a safe state and allows the system to wake using the button. After changes to interrupt handling, perform several cycles: startup – operation – sleep – wake-up.